

Upgrading custom world to 21w43a ignores min_y and height

MC-239900 Custom Worlds

Status	Resolved / Fixed
Affected	21w43a
Fixed in	1.18 Pre-release 6
Reported	2021-10-27

STEPS TO REPRODUCE

1. Launch Minecraft in version 21w42a.
2. Create a world using the custom world settings from `21w42a-height4064.json`.
3. Enter spectator mode in the created world.
4. Observe that the world extends below Y=-64 and above Y=320.
5. Launch Minecraft in version 21w43a and play the previously created world to upgrade it.
6. Observe that terrain below Y=-64 and blocks above Y=320 have been removed, with the world bottom now at Y=-64 and the world top at Y=320.

OBSERVED BEHAVIOUR

Actual behaviour

When the world created in **21w42a** is opened and upgraded in **21w43a**, the custom dimension height settings are not preserved. The dimension is loaded using the default build limits instead of the values defined in the world's custom settings:

- The effective minimum Y is changed to **-64**, so terrain and blocks generated below Y=-64 disappear.
- The effective maximum Y is changed to **320**, so blocks generated or placed above Y=320 disappear.
- The dimension's original `min_y` and height` values are therefore ignored during the upgrade.`

The affected world may also fail to load individual chunks, producing an error such as:

```
```text
```

```
Couldn't load chunk [11, -4]
```

```
java.lang.ArrayIndexOutOfBoundsException: Index 58 out of bounds for length 24
```

```
at cow.a(SourceFile:350)
```

```
at cow.a(SourceFile:290)
```

```
at cps.a(SourceFile:202)
```

```
at abu.m(SourceFile:579)
```

```
...
```

```
```
```

This results in permanent loss of terrain outside the default Y=-64 to Y=320 range after the world has been opened in 21w43a.

EXPECTED BEHAVIOUR

After upgrading the custom world to 21w43a, all previously generated terrain below Y=-64 and blocks placed above Y=320 should remain intact. The dimension's configured `min_y` and height` values should be preserved, so the world should continue to support the same vertical range as in 21w42a without removing terrain, blocks, or causing chunk-loading errors.`

ENVIRONMENT

Minecraft Java Edition Version 21w43a (snapshot)

ORIGINAL DESCRIPTION

The bug

If you open this custom world 21w42a-height4064.json in 21w43a after you generated it in 21w42a results in all terrain below -64 disappearing and any blocks being placed above 320 / max_y.

How to reproduce

- Open Minecraft in 21w42a
- Create a world using the custom world settings in 21w42a-height4064.json

Enter spectator mode, and observe that the world bottom is lower than Y=-64 and world top is higher than Y=320 (e.g. by placing a block above it)

- Open Minecraft in 21w43a
- Play the world you created previously to upgrade it

The terrain previously generated below Y=-64 has been removed, and the bottom of the world is now found at Y=-64.

Blocks above Y=320 are removed, and the top of the world is now found at Y=320

Expected behaviour

- The upgraded world would not have all terrain below Y=-64 and above Y=320 removed, the dimension's bounding boxes / min_y and height should stay unchanged.

Error

Couldn't load chunk [11, -4]

java.lang.ArrayIndexOutOfBoundsException: Index 58 out of bounds for length 24

at cow.a(SourceFile:350)

at cow.a(SourceFile:290)

at cps.a(SourceFile:202)

at abu.m(SourceFile:579)

at java.base/java.util.concurrent.CompletableFuture\$AsyncSupply.run(CompletableFuture.java:1764)

at atg.c(SourceFile:151)

at acf\$b.c(SourceFile:534)

at atg.y(SourceFile:125)

at acf\$b.y(SourceFile:543)

at acf.d(SourceFile:278)

at net.minecraft.server.MinecraftServer.be(SourceFile:763)

at net.minecraft.server.MinecraftServer.y(SourceFile:751)

at atg.c(SourceFile:134)

at net.minecraft.server.MinecraftServer.x(SourceFile:736)

at net.minecraft.server.MinecraftServer.b(SourceFile:497)

at net.minecraft.server.MinecraftServer.f_(SourceFile:329)

at fcg.e(SourceFile:72)

at net.minecraft.server.MinecraftServer.w(SourceFile:655)

at net.minecraft.server.MinecraftServer.a(SourceFile:269)

at java.base/java.lang.Thread.run(Thread.java:831)

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-239900>